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OptSurvCutR: Continued Development, Validation, and Clinical Accessibility

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1 Executive Summary

Survival analysis often requires splitting a continuous predictor, such as a biomarker or a lab value, into risk groups. The usual methods, a median split or a search for the single most significant cut-point, do not check how stable the threshold is under resampling, and the selection process itself inflates apparent significance.

OptSurvCutR is an R package, available on CRAN, that addresses this. It searches for cut-points while adjusting for covariates, supports more than two risk tiers, and uses bootstrap resampling to give each result an automated stability grade before it is reported. The package is under rOpenSci software review, and has already been used independently by a genomics group on UK Biobank and All of Us data.

This proposal is a resubmission, narrowed in scope after informal feedback. It funds the next phase in three milestones over six months:

1. **Simulation study:** test, on data with known thresholds, whether the stability tiers predict recovery of the true threshold out of sample. A null result will be reported as a finding.
2. **Preprint update and journal submission:** add the simulation results and rOpenSci outcomes, and submit to *BMC Bioinformatics* (submission, not acceptance).
3. **Clinician-runnable script and rOpenSci completion:** one script that runs the three-step workflow on a clinician's own data and produces a plain-language summary and standard plots, delivered alongside the remaining review actions.

Team and budget: Payton (Tung On) Yau, author and maintainer, and Suhirthakumar Puvanendran, co-author, each contribute about 2 to 3 hours a week. We request \$2,100, all of it labour (about 130 hours in total), divided equally across the three milestones.

2 Signatories

2.1 Project team

Payton (Tung On) Yau - author and maintainer of `OptSurvCutR` (CRAN, DOI: [10.32614/CRAN.package.OptSurvCutR](https://doi.org/10.32614/CRAN.package.OptSurvCutR)), lead on this proposal's delivery. Biomedical researcher (PhD by published work in Medicine, NTU 2023) with a combined wet-lab and computational background; currently an hourly-paid lecturer at Nottingham Trent University. github.com/paytonyau/OptSurvCutR · ORCID [0000-0002-3283-0370](https://orcid.org/0000-0002-3283-0370)

Suhirthakumar Puvanendran - co-author of `OptSurvCutR`, actively contributing to the user experience, the four-tier stability reporting system, the simulation work, and the work this proposal funds. Registered Nurse (Adult) with a PhD in Bioinformatics (2025) · ORCID: [0000-0003-2346-0483](https://orcid.org/0000-0003-2346-0483)

2.2 Contributors

None beyond the project team.

2.3 Consulted

Josiah Parry (ISC committee member) gave informal, unofficial feedback on this project's previous submission, which shaped the scope and milestones of this resubmission (see Section 3 and Section 4). His comments do not imply ISC endorsement.

The rOpenSci review team: the package is under rOpenSci software review ([review issue](#)), and we are aiming to meet its Gold standard.

2.4 Track record

`OptSurvCutR` has already been used independently. A genomics group at CHU Sainte-Justine (Montreal, Canada) used its genetic search to select a polygenic-risk-score cut-point in the UK Biobank exome cohort ($n = 493,659$) for a 2026 medRxiv preprint on genetic risk factors for immune cytopenia ([Da Silva Faria et al. 2026](#)). Analysis scripts they shared with us apply it in All of Us as well, and to a composite multi-variant score.

3 The Problem

Survival analysis often splits a continuous predictor, such as a biomarker, lab value or gene expression, into risk groups. The usual approach is a median split or a search for the single most significant cut-point. Neither checks how stable the cut-point is under resampling, and the search itself inflates apparent significance, so a threshold can look convincing in one cohort and misclassify patients in the next. This affects prognostic biomarker stratification in any field where continuous predictors and time-to-event outcomes co-occur. Solving it lets researchers report a threshold with evidence of how well it holds up, not a p-value alone.

Existing R packages that search for a cut-point, such as `survminer` ([Kassambara et al. 2024](#)) and `maxstat` ([Hothorn 2017](#)), find a single statistically significant threshold. They do not adjust for covariates or check its stability as a built-in step. Payton ran into this limit during his MSc and again in later industrial work: a single tipping point assumes risk only rises or falls in one direction, which biology often does not. A biomarker that is risky at both low and high levels, for example, has no single threshold that captures it. `OptSurvCutR` ([Yau and Puvanendran 2026](#)) was built to close

that gap: it adjusts for covariates, supports multiple risk tiers rather than forcing a binary split, and uses bootstrap resampling to assign an automated stability grade before a result is reported.

4 The proposal

4.1 Overview

Writing the code was the easy part. Validating it, maintaining it, and earning the community’s trust to become a standard is the ongoing work this proposal supports.

OptSurvCutR was not yet on CRAN when we first applied; it is now (v0.11.1). This grant would pay for what comes next. First, a simulation study to test whether the stability grades mean what we say they mean. Second, an update to the bioRxiv preprint (Yau and Puvanendran 2025) so we can submit to *BMC Bioinformatics*. Third, a simple script for researchers who are not R specialists, delivered while the rOpenSci review finishes. The plan runs over six months (Section 5).

4.2 Detail

4.2.1 Minimum Viable Product

The smallest complete deliverable is the simulation study. The manuscript is explicit about why this matters. It states that bootstrap resampling does not establish that a threshold will be recovered in an independent population, that this relationship has not been evaluated, and that the tier boundaries of 30% and 60% are “pragmatic conventions rather than calibrated thresholds.” The simulation study is designed to test exactly that.

4.2.2 Architecture

Both deliverables sit on top of the existing package: the simulation study is a self-contained Quarto document that tests its outputs against known ground truth, and the script wraps `find_cutpoint_number()`, `find_cutpoint()` and `validate_cutpoint()` without changing the statistical engines.

4.2.3 Assumptions

- The simulation finishes on time. If the tiers turn out not to calibrate, we will report that as a finding.
- The target user for the script can run an R script but does not want to learn the full function-level API; if not, that milestone is re-scoped toward documentation.
- rOpenSci review continues on a compatible timeline; if it stalls, we report progress rather than completion.

4.2.4 External dependencies

survival, survminer, rgenoud, Rcpp, ggplot2, patchwork, cli, tidyr, rlang, foreach, doParallel and doRNG (the package’s Imports at v0.11.1). No new dependencies are anticipated.

5 Project plan

5.1 Start-up phase

Start-up is largely complete because the project is already in motion. The code is public on GitHub with issue tracking open, releases are documented in `NEWS.md`, and milestone updates will be posted to blogs. Licensing is described under Other aspects.

5.2 Technical delivery

Milestone 1 - Simulation study (target: month 1-3; a pilot run is done, the final study has not started). Data simulated under known one-cut and two-cut ground truths, across multiple sample sizes and event rates, testing whether the package’s stability tiers predict out-of-sample recovery of the true threshold. Deliverable: public simulation code, results, and a written report.

Milestone 2 - Preprint update and journal submission (target: month 3-5). Update the bioRxiv preprint to incorporate Milestone 1’s results and any rOpenSci review outcomes. Deliverable: updated preprint and submitted manuscript.

Milestone 3 - Clinician-runnable script and rOpenSci completion (target: month 3-6, ongoing throughout). A single R script (or Quarto document) that takes a clinician’s own dataset (one row per patient: predictor, follow-up time, event status, optional covariates), runs the package’s three-step pipeline, and produces a plain-language summary and standard plots, configured by naming a handful of columns. Deliverable: script, short usage guide and worked PBC example; rOpenSci acceptance or a clear remaining action list.

5.3 Other aspects

Open-source under GPL-3 (package) and CC-BY (documentation and vignettes). Publicity plan:

- **Announcement post** for this grant, on acceptance
- **Delivery blog posts** at months 3, 5 and 6, so progress is visible incrementally
- **Social media** at each milestone
- **Talks:** given at FOSDEM 2026 and R!sk 2026; an abstract is submitted to the 3rd International Online Conference on Genes (2 to 4 December 2026), acceptance pending.
- **ISC meetings:** available to attend or give a progress update if invited.
- **User feedback:** when we learn of public use of the package (citations, preprints, GitHub issues), we will ask the authors for brief feedback and, with permission, list the use case on a public users page. This will test the assumption behind Milestone 3 that users want a simpler entry point.

5.4 Budget & funding plan

Milestone	Work	Requested amount
1 - Simulation study	Design, run and report	\$700
2 - Preprint and submission	Update preprint, submit to <i>BMC Bioinformatics</i>	\$700
3 - Script and rOpenSci	Build script; respond to review	\$700
Total		\$2,100

At 2 to 3 hours a week each over six months, that is about 52.5 hours each, or about 105 hours in total, costed at about \$20 per hour. No other costs are included.

6 Success

6.1 Definition of done

- **Milestone 1:** the simulation report is public, including any failure of the tiers to calibrate.
- **Milestone 2:** the manuscript is submitted to *BMC Bioinformatics* and the preprint updated (submission, not acceptance).
- **Milestone 3:** the script runs on the PBC example and one independent dataset, and `rOpenSci` review reaches a decision or a documented list of remaining actions by month 6.

6.2 Measuring success

- CRAN download trend (via `cranlogs`), as a proxy for uptake
- External issues and pull requests, as a proxy for use beyond the authors
- The `rOpenSci` review thread, a public marker that does not rely on our own reporting
- Simulation results, whatever they show: a null result is still an honest answer to a testable question
- Feedback exchanges with users, and use cases listed with permission

6.3 Future work

The method is not specific to medicine; any time-to-event problem with a continuous covariate could use it, such as turbine component failure against vibration load, or tree mortality against soil moisture. Reaching those fields needs domain-specific examples and engagement with those communities, so it is a direction the design stays open to, not part of this grant. Nearer extensions, also out of scope, are a decision-consequence simulation that attaches a modelled treatment effect to the risk groups, and a Shiny interface for users who do not write R.

Da Silva Faria, Stennio, Julie Bineau, Rafael Moisan, Marc-André Legault, Estelle Lecluze, and Thomas Pincez. 2026. *Rare and Common Germline and Somatic Variants Shape Immune Cytopenia Risk and Enable Risk Stratification*. medRxiv. <https://doi.org/10.64898/2026.08.26.26361484>.

Hothorn, Torsten. 2017. *Maxstat: Maximally Selected Rank Statistics*. <https://CRAN.R-project.org/package=maxstat>.

Kassambara, Alboukadel, Marcin Kosinski, and Przemyslaw Biecek. 2024. *Survminer: Drawing Survival Curves Using 'Ggplot2'*. <https://CRAN.R-project.org/package=survminer>.

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Yau, Payton, and Suhirthakumar Puvanendran. 2026. *OptSurvCutR: Optimal Survival Cut-Point Discovery for Time-to-Event Analysis*. <https://doi.org/10.32614/CRAN.package.OptSurvCutR>.